Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Fagundes Bros. Dairy #3

Physical address of dairy:

Number and Street	City	County	Zip Code
2852 Turlock RD	Snelling	Merced	95369

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1954

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X042-X200-X058-XXXX X042-X200-X059-XXXX

B. OPERATORS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Operator name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This operator is responsible for paying permit fees.

C. OWNERS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Legal owner name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	1,063	108	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	1,063	108	0	0	0	0
Average number	1,063	108	0	0	0	0
Avg live weight (lbs)	1,300	1,400	0	0		

Predominant milk cow breed: Holstein

Average milk production: 63 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 27,945.19 *tons per reporting period*

Total nitrogen from manure: 364,801.13 *lbs per reporting period*

After ammonia losses (30% loss applied): 255,360.79 *lbs per reporting period*

Total phosphorus from manure: 60,857.12 *lbs per reporting period*

Total potassium from manure: 198,685.57 *lbs per reporting period*

Total salt from manure: 525,348.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 45,594,000 *gallons*

Total nitrogen generated: 88,359.70 *lbs*

Total phosphorus generated: 14,939.39 *lbs*

Total potassium generated: 97,122.78 *lbs*

Total salt generated: 592,456.78 *lbs*

	45,594,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	45,594,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DW 1	Ground water
Domestic Well - DW 2	Ground water
Domestic Well - DW 3	Ground water
Irrigation Water - Surface Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
26	15	15	1	manure	X042-X200-X059-XXXX
28	15	15	1	manure	X042-X200-X059-XXXX
45	52	52	3	process wastewater	X042-X200-X050-XXXX
46	90	90	2	process wastewater	X042-X200-X052-XXXX
47	81	81	2	process wastewater	X042-X200-X059-XXXX
Carson	30	30	2	manure	X043-X070-X005-XXXX
CMI	30	30	1	manure	X042-X200-X046-XXXX
Collins	58	58	2	manure	X043-X007-X038-XXXX
Field 44	45	45	1	process wastewater	X042-X200-X059-XXXX
Silvera 1	72	72	2	manure	X043-X007-X004-XXXX
Silvera 2	30	30	2	manure	X043-X007-X004-XXXX
Silvera 3	39	39	2	manure	X043-X070-X004-XXXX
Totals for areas that were used for application	557	557	21		
Totals for areas that were not used for application					
Land application area totals	557	557	21		

B. CROPS AND HARVESTS

26

Field name: 26

11/20/2020: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 15 Plant date: 11/20/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	31.00 *ton*	Dry-weight		62.5	31,500.00	2,900.00	24,700.00		16.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	2.07	48.83	4.50	38.29	260.40

28

28

Field name: 28

11/20/2020: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 15 Plant date: 11/20/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	32.00 *ton*	Dry-weight		62.5	31,500.00	2,900.00	24,700.00		16.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	2.13	50.40	4.64	39.52	268.80

45

Field name: 45

12/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 12/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	749.00 *ton*	Dry-weight		68.3	16,900.00	3,100.00	20,000.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.40	154.33	28.31	182.64	484.00

05/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 05/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	1,398.00 *ton*	Dry-weight		70.7	15,100.00	2,800.00	19,500.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.88	237.89	44.11	307.21	1,134.32

45

09/14/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 52 Plant date: 09/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	612.00 *ton*	Dry-weight		83.8	20,400.00	3,300.00	31,600.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	11.77	77.79	12.58	120.50	411.83

46

Field name: 46

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 90 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	1,479.00 *ton*	Dry-weight		70.9	15,900.00	2,000.00	17,100.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.43	152.07	19.13	163.55	765.14

05/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 90 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/13/2023	1,061.00 *ton*	Dry-weight		77.1	14,800.00	270.00	31,900.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	11.79	79.91	1.46	172.24	512.93

47

Field name: 47

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 81 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	1,022.00 *ton*	Dry-weight		73.1	22,400.00	4,000.00	15,200.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.62	152.05	27.15	103.18	488.74

05/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 81 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/13/2023	843.00 *ton*	Dry-weight		77.5	17,600.00	360.00	40,900.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.41	82.43	1.69	191.55	552.63

Carson

Field name: Carson

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	509.00 *ton*	Dry-weight		69.1	10,700.00	2,300.00	15,600.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.97	112.19	24.12	163.57	587.18

Carson

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	963.00 *ton*	Dry-weight		63.3	8,900.00	2,600.00	14,400.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.10	209.70	61.26	339.28	1,555.05

CMI

Field name: CMI

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	475.00 *ton*	Dry-weight		70.4	15,200.00	4,500.00	23,900.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.83	142.47	42.18	224.02	806.11

Collins

Field name: Collins

Collins

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 58 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	711.00 *ton*	Dry-weight		77.6	27,700.00	3,700.00	18,100.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.26	152.12	20.32	99.40	313.04

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/02/2023	1,791.00 *ton*	Dry-weight		73.7	15,900.00	4,100.00	21,400.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.88	258.26	66.59	347.59	1,331.89

Field 44

Field name: Field 44

11/20/2020: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 45 Plant date: 11/20/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	121.00 *ton*	Dry-weight		64.2	28,500.00	3,200.00	26,100.00		25.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	2.69	54.87	6.16	50.25	483.24

Silvera 1

Field name: Silvera 1

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 72 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	1,209.00 *ton*	Dry-weight		74.8	16,700.00	5,300.00	29,300.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.79	141.33	44.85	247.97	761.67

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/02/2023	2,201.00 *ton*	Dry-weight		72.1	14,000.00	4,400.00	20,000.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.57	238.81	75.05	341.16	1,586.37

Silvera 2

Field name: Silvera 2

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	509.00 *ton*	Dry-weight		75.2	15,900.00	3,600.00	23,400.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.97	133.81	30.30	196.92	639.58

Silvera 2

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/02/2023	895.00 *ton*	Dry-weight		72.0	15,100.00	5,500.00	21,700.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.83	252.27	91.89	362.53	1,637.25

Silvera 3

Field name: Silvera 3

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	617.00 *ton*	Dry-weight		75.1	17,200.00	4,000.00	25,800.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.82	135.51	31.51	203.27	646.05

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	1,279.00 *ton*	Dry-weight		74.0	11,900.00	4,000.00	23,000.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.79	202.93	68.21	392.23	1,415.43

NUTRIENT BUDGET

A. LAND APPLICATIONS

26 - 11/20/2020: Clover-grass, hay

Field name: 26

Crop: Clover-grass, hay

Plant date: 11/20/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.40	1,643,340.00 *gal*
Application event totals		0.00	0.00	0.00	38.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.34	1,769,060.00 *gal*
Application event totals		0.00	0.00	0.00	41.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.24	1,679,260.00 *gal*
Application event totals		0.00	0.00	0.00	39.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.61	1,652,320.00 *gal*
Application event totals		0.00	0.00	0.00	38.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.92	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	40.92	

26 - 11/20/2020: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.76	1,787,020.00 *gal*
Application event totals		0.00	0.00	0.00	41.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.34	1,769,060.00 *gal*
Application event totals		0.00	0.00	0.00	41.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	51.92	3.99	34.94	0.00	120.00 *ton*
Application event totals		51.92	3.99	34.94	0.00	

28 - 11/20/2020: Clover-grass, hay

Field name: 28

Crop: Clover-grass, hay

Plant date: 11/20/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.98	1,625,380.00 *gal*
Application event totals		0.00	0.00	0.00	37.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.92	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	40.92	

28 - 11/20/2020: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	44.27	1,894,780.00 *gal*
Application event totals		0.00	0.00	0.00	44.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.82	1,661,300.00 *gal*
Application event totals		0.00	0.00	0.00	38.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	44.69	1,912,740.00 *gal*
Application event totals		0.00	0.00	0.00	44.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.76	1,787,020.00 *gal*
Application event totals		0.00	0.00	0.00	41.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.34	1,769,060.00 *gal*
Application event totals		0.00	0.00	0.00	41.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	56.24	4.33	37.86	0.00	130.00 *ton*
Application event totals		56.24	4.33	37.86	0.00	

45 - 12/27/2022: Oats, silage-soft dough

45 - 12/27/2022: Oats, silage-soft dough

Field name: 45

Crop: Oats, silage-soft dough

Plant date: 12/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.82	11.92	77.04	470.13	1,890,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	35.53	5,271,260.00 *gal*
Application event totals		70.82	11.92	77.04	505.66	

03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	71.27	12.00	77.53	473.11	1,902,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	35.47	5,262,280.00 *gal*
Application event totals		71.27	12.00	77.53	508.58	

03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.35	11.50	74.35	453.71	1,824,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	35.53	5,271,260.00 *gal*
Application event totals		68.35	11.50	74.35	489.24	

45 - 05/11/2023: Corn, silage

Field name: 45

Crop: Corn, silage

Plant date: 05/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	47.09	6,986,440.00 *gal*
Application event totals		0.00	0.00	0.00	47.09	

45 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.70	18.13	117.15	714.89	2,874,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	47.03	6,977,460.00 *gal*
Application event totals		107.70	18.13	117.15	761.92	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.47	18.09	116.91	713.40	2,868,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	47.03	6,977,460.00 *gal*
Application event totals		107.47	18.09	116.91	760.43	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	47.15	6,995,420.00 *gal*
Application event totals		0.00	0.00	0.00	47.15	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.25	18.05	116.66	711.91	2,862,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	46.79	6,941,540.00 *gal*
Application event totals		107.25	18.05	116.66	758.70	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	46.79	6,941,540.00 *gal*
Application event totals		0.00	0.00	0.00	46.79	

45 - 09/14/2023: Sudangrass, silage

Field name: 45

Crop: Sudangrass, silage Plant date: 09/14/2023

45 - 09/14/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.53	9.35	60.41	368.64	1,482,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.68	5,738,220.00 *gal*
Application event totals		55.53	9.35	60.41	407.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.02	9.05	69.40	419.26	1,476,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.46	5,854,960.00 *gal*
Application event totals		46.02	9.05	69.40	458.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.68	5,738,220.00 *gal*
Application event totals		0.00	0.00	0.00	38.68	

46 - 11/09/2022: Oats, silage-soft dough

Field name: 46

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.68	11.39	73.62	449.27	3,126,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	36.47	9,366,140.00 *gal*
Application event totals		67.68	11.39	73.62	485.74	

46 - 11/09/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.38	11.17	72.21	440.64	3,066,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	37.59	9,653,500.00 *gal*
Application event totals		66.38	11.17	72.21	478.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.77	11.24	72.63	443.23	3,084,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.12	9,788,200.00 *gal*
Application event totals		66.77	11.24	72.63	481.35	

46 - 05/25/2023: Sudangrass, silage

Field name: 46

Crop: Sudangrass, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.57	8.68	56.10	342.34	2,382,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	45.08	11,575,220.00 *gal*
Application event totals		51.57	8.68	56.10	387.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.18	8.61	55.68	339.75	2,364,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	45.43	11,665,020.00 *gal*
Application event totals		51.18	8.61	55.68	385.18	

46 - 05/25/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	44.73	11,485,420.00 *gal*
Application event totals		0.00	0.00	0.00	44.73	

47 - 11/09/2022: Oats, silage-soft dough

Field name: 47

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.98	11.44	73.95	451.28	2,826,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.75	9,186,540.00 *gal*
Application event totals		67.98	11.44	73.95	491.03	

03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.56	11.54	74.58	455.11	2,850,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.30	9,545,740.00 *gal*
Application event totals		68.56	11.54	74.58	496.42	

04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.54	11.20	72.38	441.70	2,766,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	43.09	9,958,820.00 *gal*
Application event totals		66.54	11.20	72.38	484.79	

47 - 05/25/2023: Sudangrass, silage

47 - 05/25/2023: Sudangrass, silage

Field name: 47

Crop: Sudangrass, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.13	9.11	58.88	359.30	2,250,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	49.54	11,449,500.00 *gal*
Application event totals		54.13	9.11	58.88	408.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.42	9.16	59.19	361.21	2,262,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	50.16	11,593,180.00 *gal*
Application event totals		54.42	9.16	59.19	411.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	49.15	11,359,700.00 *gal*
Application event totals		0.00	0.00	0.00	49.15	

Carson - 11/09/2022: Oats, silage-soft dough

Field name: Carson

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.65	3,565,060.00 *gal*
Application event totals		0.00	0.00	0.00	41.65	

Carson - 11/09/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.71	3,484,240.00 *gal*
Application event totals		0.00	0.00	0.00	40.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.39	3,457,300.00 *gal*
Application event totals		0.00	0.00	0.00	40.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	147.84	28.41	69.08	0.00	360.00 *ton*
Application event totals		147.84	28.41	69.08	0.00	

Carson - 05/27/2023: Corn, silage

Field name: Carson

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	81.52	6,977,460.00 *gal*
Application event totals		0.00	0.00	0.00	81.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	287.47	55.23	134.32	0.00	700.00 *ton*
Application event totals		287.47	55.23	134.32	0.00	

Carson - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	80.47	6,887,660.00 *gal*
Application event totals		0.00	0.00	0.00	80.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	79.84	6,833,780.00 *gal*
Application event totals		0.00	0.00	0.00	79.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	80.05	6,851,740.00 *gal*
Application event totals		0.00	0.00	0.00	80.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	78.58	6,726,020.00 *gal*
Application event totals		0.00	0.00	0.00	78.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	79.00	6,761,940.00 *gal*
Application event totals		0.00	0.00	0.00	79.00	

CMI - 11/04/2022: Oats, silage-soft dough

Field name: CMI

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

CMI - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.65	3,565,060.00 *gal*
Application event totals		0.00	0.00	0.00	41.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.81	3,493,220.00 *gal*
Application event totals		0.00	0.00	0.00	40.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.96	3,505,792.00 *gal*
Application event totals		0.00	0.00	0.00	40.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	184.80	35.51	86.35	0.00	450.00 *ton*
Application event totals		184.80	35.51	86.35	0.00	

Collins - 10/24/2022: Oats, silage-soft dough

Field name: Collins

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.84	6,923,580.00 *gal*
Application event totals		0.00	0.00	0.00	41.84	

Collins - 10/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.24	6,824,800.00 *gal*
Application event totals		0.00	0.00	0.00	41.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.24	6,824,800.00 *gal*
Application event totals		0.00	0.00	0.00	41.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	203.92	39.18	95.28	0.00	960.00 *ton*
Application event totals		203.92	39.18	95.28	0.00	

Collins - 05/10/2023: Corn, silage

Field name: Collins

Crop: Corn, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	350.49	67.34	163.77	0.00	1,650.00 *ton*
Application event totals		350.49	67.34	163.77	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	56.60	9,366,140.00 *gal*
Application event totals		0.00	0.00	0.00	56.60	

Collins - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	56.49	9,348,180.00 *gal*
Application event totals		0.00	0.00	0.00	56.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	56.38	9,330,220.00 *gal*
Application event totals		0.00	0.00	0.00	56.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.97	9,096,740.00 *gal*
Application event totals		0.00	0.00	0.00	54.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	53.56	8,863,260.00 *gal*
Application event totals		0.00	0.00	0.00	53.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	53.89	8,917,140.00 *gal*
Application event totals		0.00	0.00	0.00	53.89	

Field 44 - 11/20/2020: Clover-grass, hay

Field name: Field 44

Crop: Clover-grass, hay

Plant date: 11/20/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 44 - 11/20/2020: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	38.96	5,001,860.00 *gal*
Application event totals		0.00	0.00	0.00	38.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.66	5,091,660.00 *gal*
Application event totals		0.00	0.00	0.00	39.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.10	5,019,820.00 *gal*
Application event totals		0.00	0.00	0.00	39.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.35	10.49	67.83	413.91	1,440,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.38	5,055,740.00 *gal*
Application event totals		62.35	10.49	67.83	453.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	40.15	5,154,520.00 *gal*
Application event totals		0.00	0.00	0.00	40.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.80	5,109,620.00 *gal*
Application event totals		0.00	0.00	0.00	39.80	

Field 44 - 11/20/2020: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.94	5,127,580.00 *gal*
Application event totals		0.00	0.00	0.00	39.94	

Silvera 1 - 11/04/2022: Oats, silage-soft dough

Field name: Silvera 1

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.47	8,108,940.00 *gal*
Application event totals		0.00	0.00	0.00	39.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	41.13	8,450,180.00 *gal*
Application event totals		0.00	0.00	0.00	41.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.80	8,791,420.00 *gal*
Application event totals		0.00	0.00	0.00	42.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	188.23	36.17	87.95	0.00	1,100.00 *ton*
Application event totals		188.23	36.17	87.95	0.00	

Silvera 1 - 05/10/2023: Corn, silage

Field name: Silvera 1

Crop: Corn, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	325.12	62.47	151.91	0.00	1,900.00 *ton*
Application event totals		325.12	62.47	151.91	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	52.41	10,767,020.00 *gal*
Application event totals		0.00	0.00	0.00	52.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	52.33	10,749,060.00 *gal*
Application event totals		0.00	0.00	0.00	52.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	52.37	10,758,040.00 *gal*
Application event totals		0.00	0.00	0.00	52.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	51.71	10,623,340.00 *gal*
Application event totals		0.00	0.00	0.00	51.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	51.80	10,641,300.00 *gal*
Application event totals		0.00	0.00	0.00	51.80	

Silvera 1 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	51.84	10,650,280.00 *gal*
Application event totals		0.00	0.00	0.00	51.84	

Silvera 2 - 11/04/2022: Oats, silage-soft dough

Field name: Silvera 2

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.34	3,367,500.00 *gal*
Application event totals		0.00	0.00	0.00	39.34	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.55	3,385,460.00 *gal*
Application event totals		0.00	0.00	0.00	39.55	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	39.97	3,421,380.00 *gal*
Application event totals		0.00	0.00	0.00	39.97	

11/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	180.70	34.72	84.43	0.00	440.00 *ton*
Application event totals		180.70	34.72	84.43	0.00	

Silvera 2 - 05/10/2023: Corn, silage

Field name: Silvera 2

Crop: Corn, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	344.97	66.28	161.18	0.00	840.00 *ton*
Application event totals		344.97	66.28	161.18	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	56.55	4,840,220.00 *gal*
Application event totals		0.00	0.00	0.00	56.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	56.13	4,804,300.00 *gal*
Application event totals		0.00	0.00	0.00	56.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.66	4,678,580.00 *gal*
Application event totals		0.00	0.00	0.00	54.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.18	4,723,480.00 *gal*
Application event totals		0.00	0.00	0.00	55.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.29	4,732,460.00 *gal*
Application event totals		0.00	0.00	0.00	55.29	

Silvera 2 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.08	4,714,500.00 *gal*
Application event totals		0.00	0.00	0.00	55.08	

Silvera 3 - 11/04/2022: Oats, silage-soft dough

Field name: Silvera 3

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	30.26	3,367,500.00 *gal*
Application event totals		0.00	0.00	0.00	30.26	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	32.36	3,600,980.00 *gal*
Application event totals		0.00	0.00	0.00	32.36	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	34.14	3,798,540.00 *gal*
Application event totals		0.00	0.00	0.00	34.14	

11/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	183.22	35.20	85.61	0.00	580.00 *ton*
Application event totals		183.22	35.20	85.61	0.00	

Silvera 3 - 05/27/2023: Corn, silage

Field name: Silvera 3

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	44.31	4,930,020.00 *gal*
Application event totals		0.00	0.00	0.00	44.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	277.99	53.41	129.89	0.00	880.00 *ton*
Application event totals		277.99	53.41	129.89	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	44.06	4,903,080.00 *gal*
Application event totals		0.00	0.00	0.00	44.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	44.06	4,903,080.00 *gal*
Application event totals		0.00	0.00	0.00	44.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	43.01	4,786,340.00 *gal*
Application event totals		0.00	0.00	0.00	43.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.85	4,768,380.00 *gal*
Application event totals		0.00	0.00	0.00	42.85	

Silvera 3 - 05/27/2023: Corn, silage

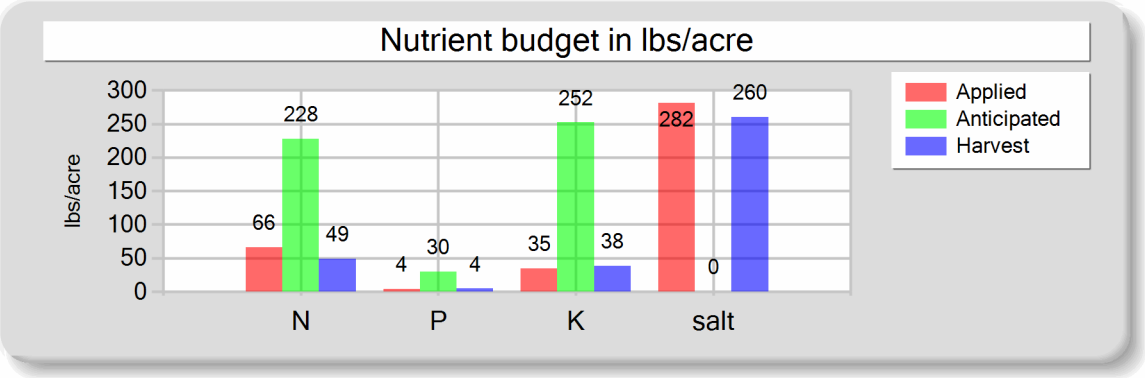
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	42.05	4,678,580.00 *gal*
Application event totals		0.00	0.00	0.00	42.05	

B. NUTRIENT BUDGET

26 - 11/20/2020: Clover-grass, hay

Field name: 26 Crop: Clover-grass, hay Plant date: 11/20/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	51.92	3.99	34.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	281.59
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	65.92	3.99	34.94	281.59
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	48.83	4.50	38.29	260.40
Nutrient balance	17.09	-0.50	-3.34	21.19
Applied to removed ratio	1.35	0.89	0.91	1.08

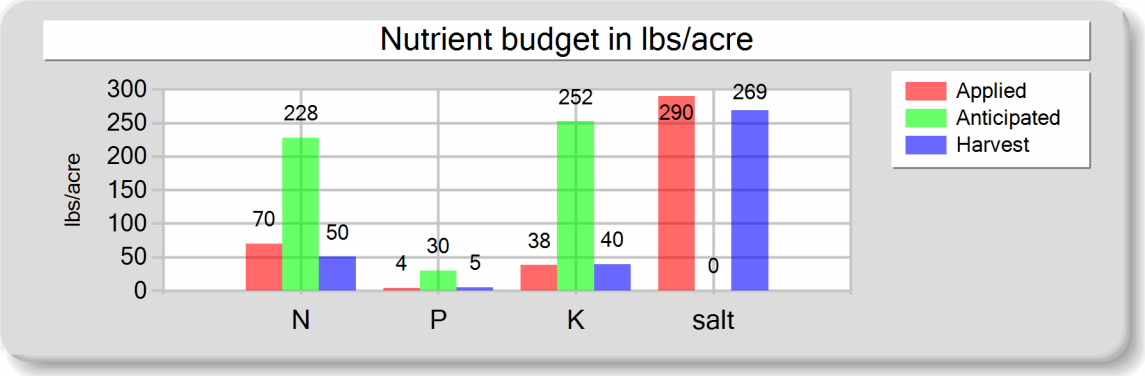
Fresh water applied	
12,051,160.00	*gallons*
443.80	*acre-inches*
29.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

28 - 11/20/2020: Clover-grass, hay

Field name: 28 Crop: Clover-grass, hay Plant date: 11/20/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	56.24	4.33	37.86	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	289.77
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	70.24	4.33	37.86	289.77
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	50.40	4.64	39.52	268.80
Nutrient balance	19.84	-0.31	-1.66	20.97
Applied to removed ratio	1.39	0.93	0.96	1.08

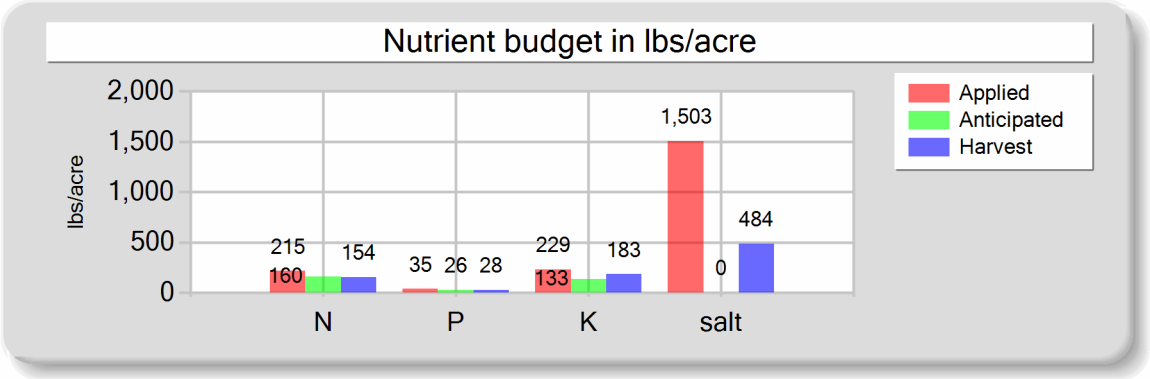
Fresh water applied
12,401,380.00 *gallons*
456.70 *acre-inches*
30.45 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

45 - 12/27/2022: Oats, silage-soft dough

Field name: 45 Crop: Oats, silage-soft dough Plant date: 12/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.44	35.42	228.92	1,396.95
Fresh water	0.00	0.00	0.00	106.53
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	215.11	35.42	228.92	1,503.48
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.33	28.31	182.64	484.00
Nutrient balance	60.78	7.11	46.28	1,019.48
Applied to removed ratio	1.39	1.25	1.25	3.11

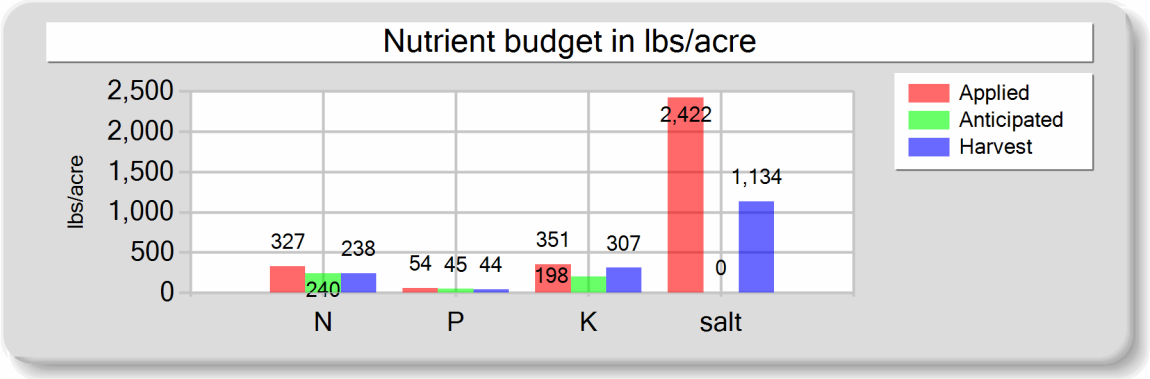
Fresh water applied	
15,804,800.00	*gallons*
582.04	*acre-inches*
11.19	*inches/acre*

Process wastewater applied	
5,616,000.00	*gallons*
206.82	*acre-inches*
3.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

45 - 05/11/2023: Corn, silage

Field name: 45 Crop: Corn, silage Plant date: 05/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	322.41	54.26	350.72	2,140.20
Fresh water	0.00	0.00	0.00	281.87
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	327.08	54.26	350.72	2,422.08
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	237.89	44.11	307.21	1,134.32
Nutrient balance	89.19	10.15	43.51	1,287.76
Applied to removed ratio	1.37	1.23	1.14	2.14

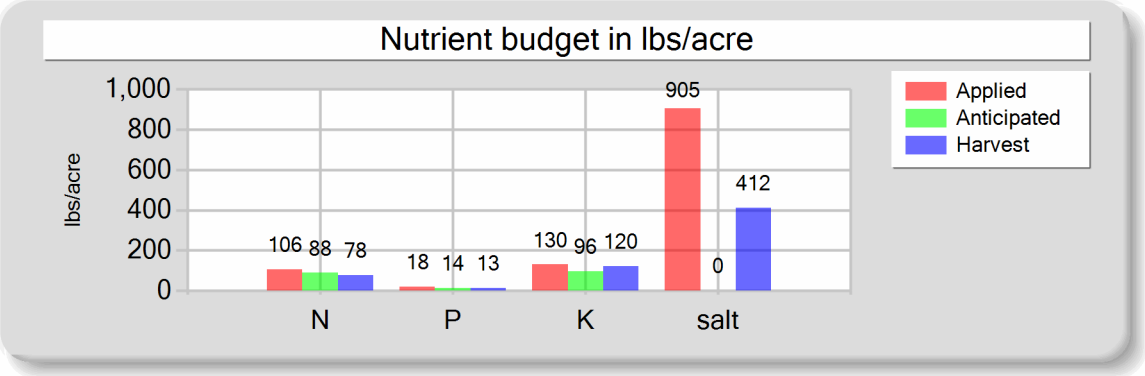
Fresh water applied	
41,819,860.00	*gallons*
1,540.08	*acre-inches*
29.62	*inches/acre*

Process wastewater applied	
8,604,000.00	*gallons*
316.86	*acre-inches*
6.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

45 - 09/14/2023: Sudangrass, silage

Field name: 45 Crop: Sudangrass, silage Plant date: 09/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	101.56	18.40	129.81	787.90
Fresh water	0.00	0.00	0.00	116.82
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	106.22	18.40	129.81	904.72
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	77.79	12.58	120.50	411.83
Nutrient balance	28.43	5.81	9.31	492.89
Applied to removed ratio	1.37	1.46	1.08	2.20

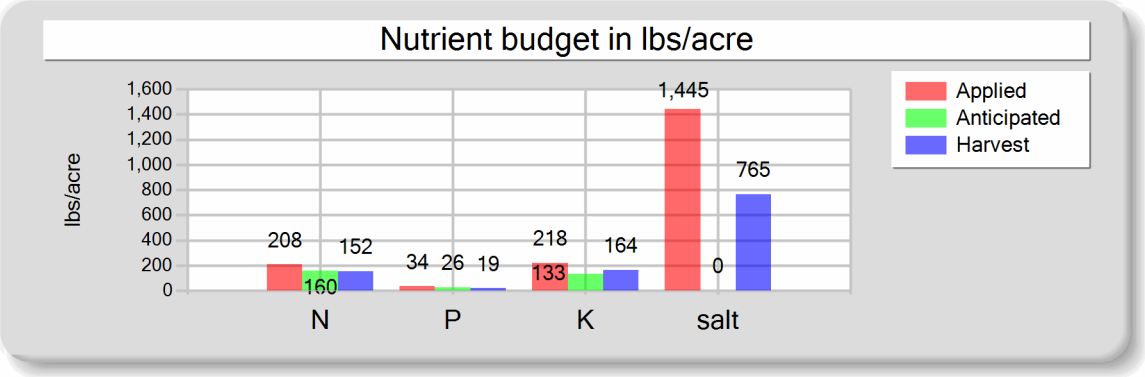
Fresh water applied	
17,331,400.00	*gallons*
638.26	*acre-inches*
12.27	*inches/acre*

Process wastewater applied	
2,958,000.00	*gallons*
108.93	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

46 - 11/09/2022: Oats, silage-soft dough

Field name: 46 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	200.83	33.80	218.46	1,333.14
Fresh water	0.00	0.00	0.00	112.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.83	33.80	218.46	1,445.33
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.07	19.13	163.55	765.14
Nutrient balance	55.76	14.67	54.92	680.19
Applied to removed ratio	1.37	1.77	1.34	1.89

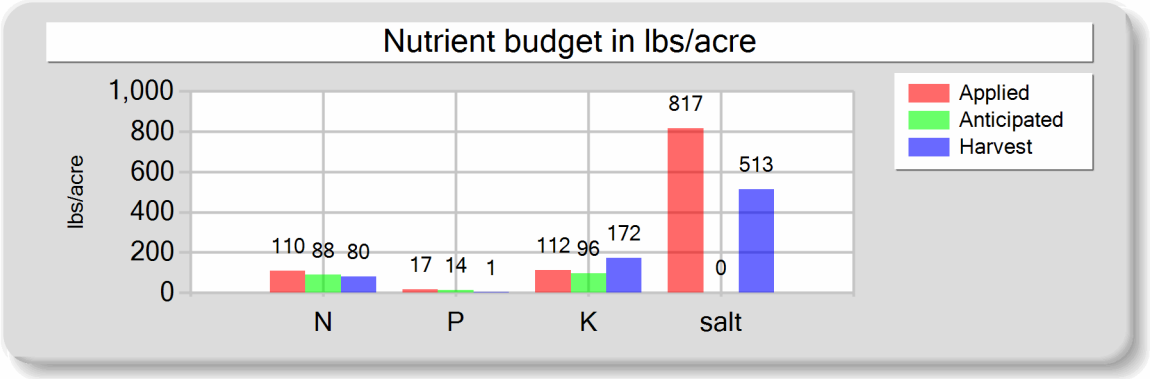
Fresh water applied	
28,807,840.00	*gallons*
1,060.89	*acre-inches*
11.79	*inches/acre*

Process wastewater applied	
9,276,000.00	*gallons*
341.60	*acre-inches*
3.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

46 - 05/25/2023: Sudangrass, silage

Field name: 46 Crop: Sudangrass, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	102.75	17.29	111.78	682.09
Fresh water	0.00	0.00	0.00	135.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	109.75	17.29	111.78	817.33
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	79.91	1.46	172.24	512.93
Nutrient balance	29.84	15.84	-60.46	304.39
Applied to removed ratio	1.37	11.86	0.65	1.59

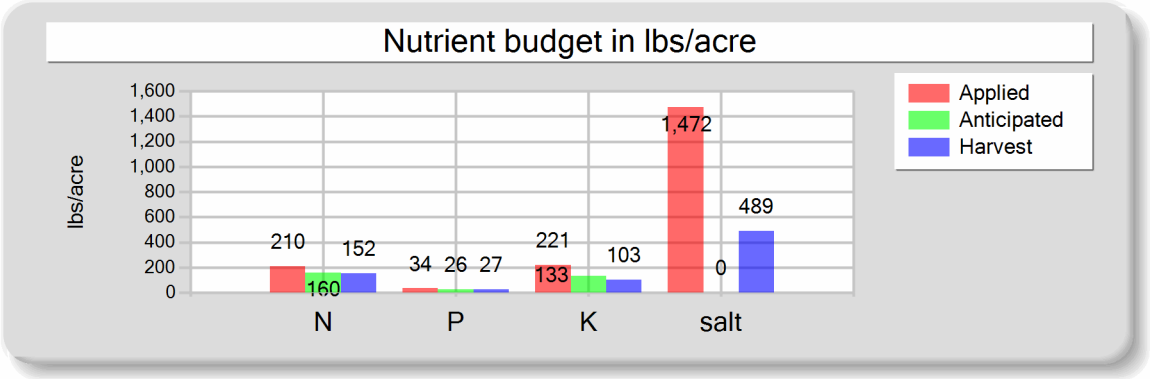
Fresh water applied	
34,725,660.00	*gallons*
1,278.83	*acre-inches*
14.21	*inches/acre*

Process wastewater applied	
4,746,000.00	*gallons*
174.78	*acre-inches*
1.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

47 - 11/09/2022: Oats, silage-soft dough

Field name: 47 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.08	34.18	220.91	1,348.09
Fresh water	0.00	0.00	0.00	124.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.08	34.18	220.91	1,472.24
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.05	27.15	103.18	488.74
Nutrient balance	58.03	7.03	117.73	983.49
Applied to removed ratio	1.38	1.26	2.14	3.01

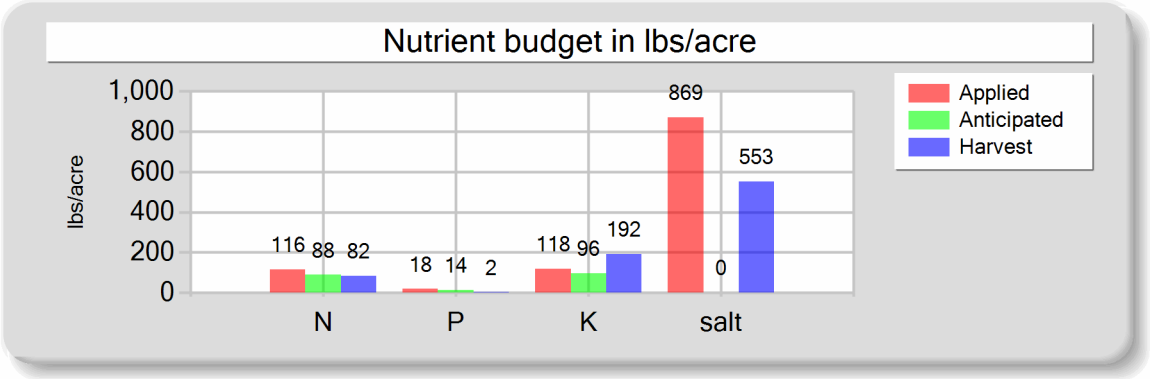
Fresh water applied	
28,691,100.00	*gallons*
1,056.60	*acre-inches*
13.04	*inches/acre*

Process wastewater applied	
8,442,000.00	*gallons*
310.89	*acre-inches*
3.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

47 - 05/25/2023: Sudangrass, silage

Field name: 47 Crop: Sudangrass, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	108.54	18.27	118.07	720.51
Fresh water	0.00	0.00	0.00	148.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	115.54	18.27	118.07	869.37
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	82.43	1.69	191.55	552.63
Nutrient balance	33.12	16.58	-73.48	316.74
Applied to removed ratio	1.40	10.84	0.62	1.57

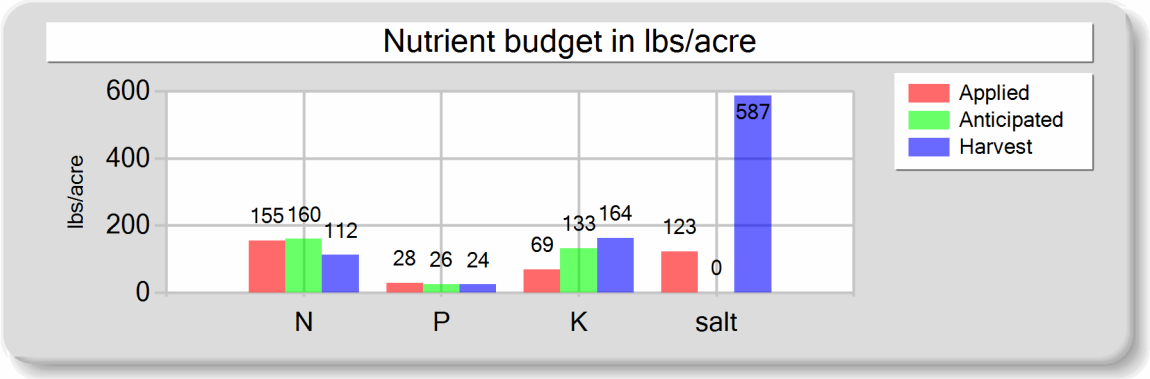
Fresh water applied	
34,402,380.00	*gallons*
1,266.92	*acre-inches*
15.64	*inches/acre*

Process wastewater applied	
4,512,000.00	*gallons*
166.16	*acre-inches*
2.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carson - 11/09/2022: Oats, silage-soft dough

Field name: Carson | Crop: Oats, silage-soft dough | Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	147.84	28.41	69.08	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	122.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.84	28.41	69.08	122.75
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.19	24.12	163.57	587.18
Nutrient balance	42.65	4.29	-94.49	-464.43
Applied to removed ratio	1.38	1.18	0.42	0.21

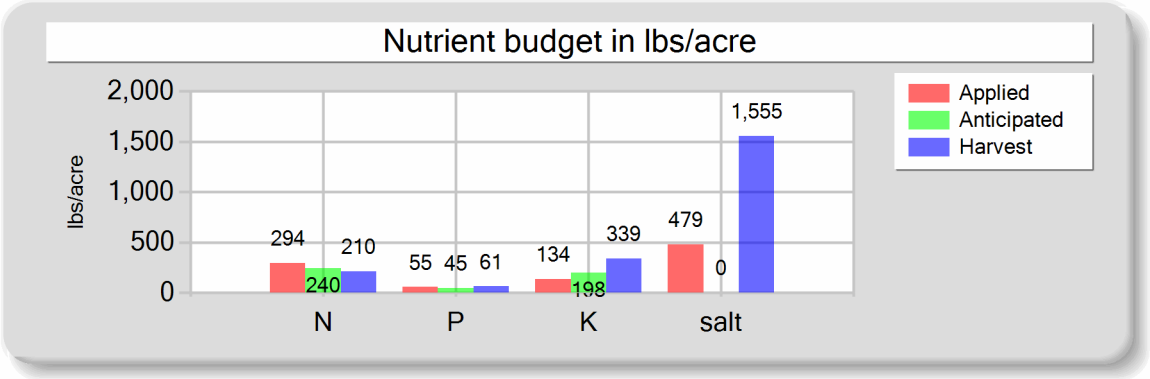
Fresh water applied	
10,506,600.00	*gallons*
386.92	*acre-inches*
12.90	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carson - 05/27/2023: Corn, silage

Field name: Carson | Crop: Corn, silage | Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	287.47	55.23	134.32	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	479.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.47	55.23	134.32	479.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.70	61.26	339.28	1,555.05
Nutrient balance	84.77	-6.02	-204.96	-1,075.60
Applied to removed ratio	1.40	0.90	0.40	0.31

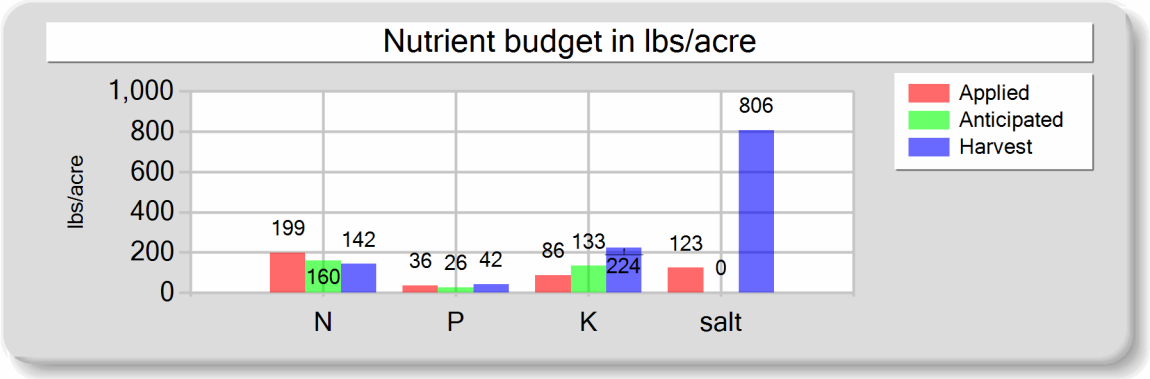
Fresh water applied	
41,038,600.00	*gallons*
1,511.31	*acre-inches*
50.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CMI - 11/04/2022: Oats, silage-soft dough

Field name: CMI | Crop: Oats, silage-soft dough | Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.80	35.51	86.35	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	123.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	198.80	35.51	86.35	123.42
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	142.47	42.18	224.02	806.11
Nutrient balance	56.33	-6.67	-137.67	-682.69
Applied to removed ratio	1.40	0.84	0.39	0.15

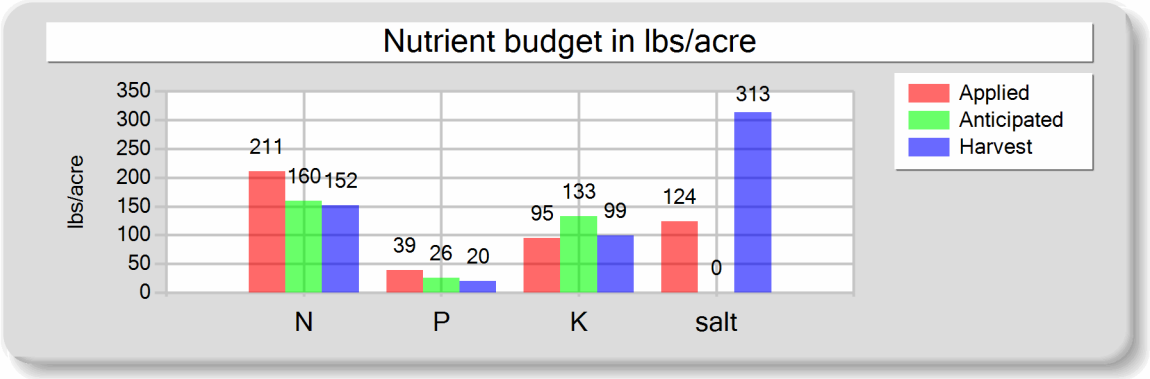
Fresh water applied	
10,564,072.00	*gallons*
389.04	*acre-inches*
12.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Collins - 10/24/2022: Oats, silage-soft dough

Field name: Collins | Crop: Oats, silage-soft dough | Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	203.92	39.18	95.28	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	124.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.92	39.18	95.28	124.32
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.12	20.32	99.40	313.04
Nutrient balance	58.80	18.86	-4.12	-188.71
Applied to removed ratio	1.39	1.93	0.96	0.40

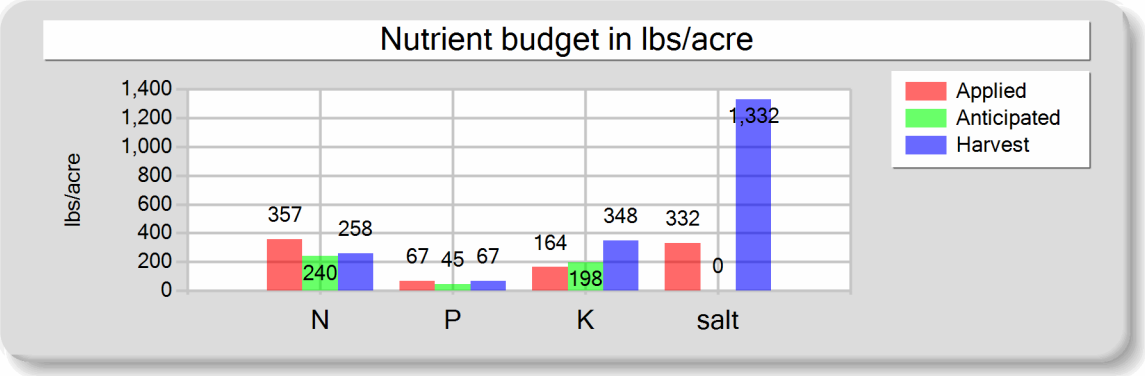
Fresh water applied	
20,573,180.00	*gallons*
757.64	*acre-inches*
13.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Collins - 05/10/2023: Corn, silage

Field name: Collins | Crop: Corn, silage | Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	350.49	67.34	163.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	331.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	357.49	67.34	163.77	331.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	258.26	66.59	347.59	1,331.89
Nutrient balance	99.23	0.75	-183.82	-1,000.00
Applied to removed ratio	1.38	1.01	0.47	0.25

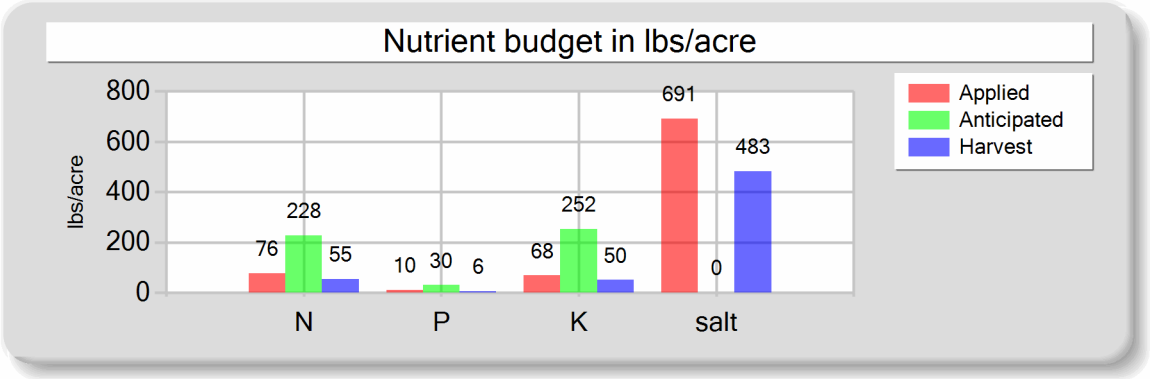
Fresh water applied	
54,921,680.00	*gallons*
2,022.58	*acre-inches*
34.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 44 - 11/20/2020: Clover-grass, hay

Field name: Field 44 Crop: Clover-grass, hay Plant date: 11/20/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	62.35	10.49	67.83	413.91
Fresh water	0.00	0.00	0.00	276.97
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	76.35	10.49	67.83	690.88
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	54.87	6.16	50.25	483.24
Nutrient balance	21.48	4.33	17.58	207.65
Applied to removed ratio	1.39	1.70	1.35	1.43

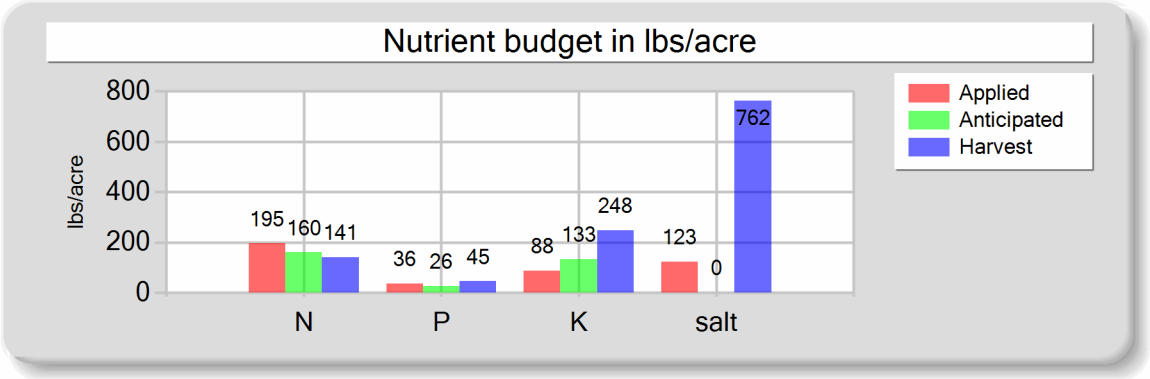
Fresh water applied	
35,560,800.00	*gallons*
1,309.58	*acre-inches*
29.10	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silvera 1 - 11/04/2022: Oats, silage-soft dough

Field name: Silvera 1 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	188.23	36.17	87.95	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	123.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.23	36.17	87.95	123.40
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.33	44.85	247.97	761.67
Nutrient balance	53.89	-8.69	-160.02	-638.27
Applied to removed ratio	1.38	0.81	0.35	0.16

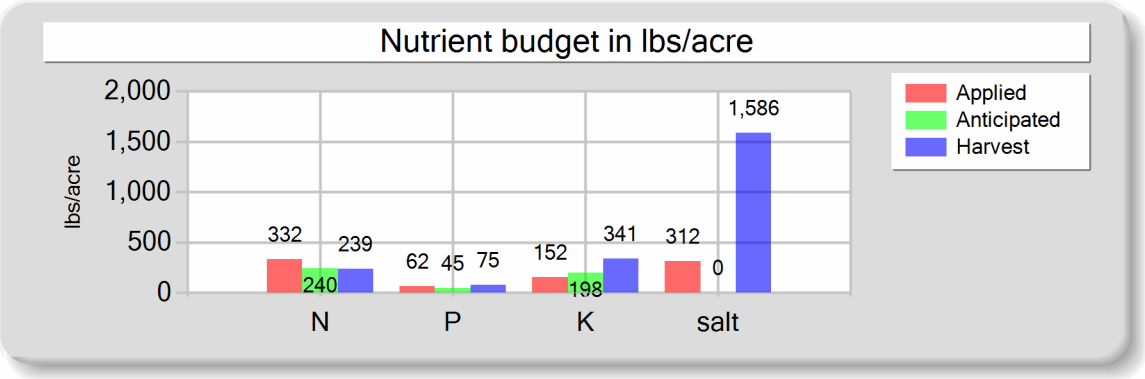
Fresh water applied	
25,350,540.00	*gallons*
933.57	*acre-inches*
12.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silvera 1 - 05/10/2023: Corn, silage

Field name: Silvera 1 Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	325.12	62.47	151.91	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	312.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	332.12	62.47	151.91	312.47
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	238.81	75.05	341.16	1,586.37
Nutrient balance	93.31	-12.59	-189.24	-1,273.90
Applied to removed ratio	1.39	0.83	0.45	0.20

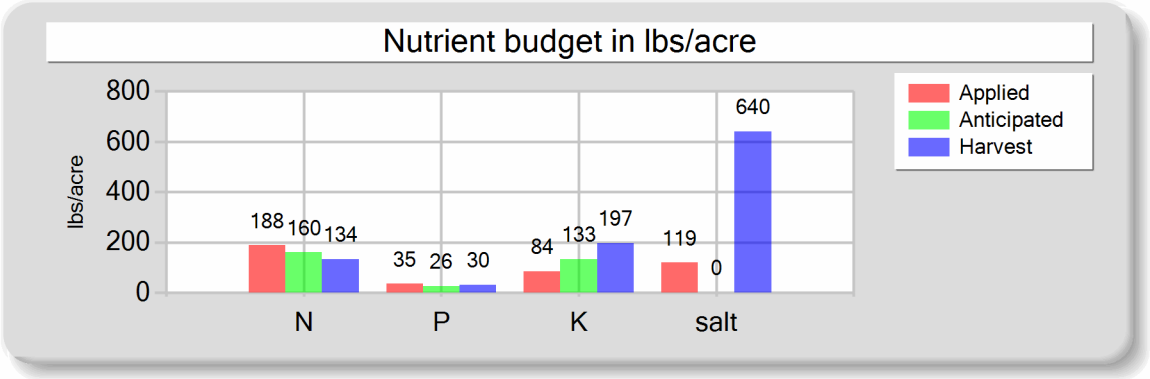
Fresh water applied	
64,189,040.00	*gallons*
2,363.86	*acre-inches*
32.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silvera 2 - 11/04/2022: Oats, silage-soft dough

Field name: Silvera 2 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	180.70	34.72	84.43	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	118.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.70	34.72	84.43	118.87
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	133.81	30.30	196.92	639.58
Nutrient balance	53.89	4.42	-112.49	-520.71
Applied to removed ratio	1.40	1.15	0.43	0.19

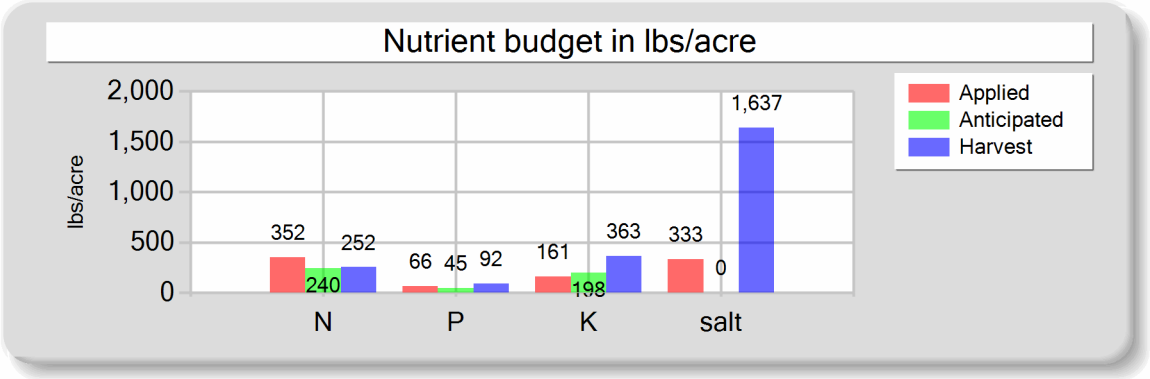
Fresh water applied	
10,174,340.00	*gallons*
374.69	*acre-inches*
12.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silvera 2 - 05/10/2023: Corn, silage

Field name: Silvera 2 Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	344.97	66.28	161.18	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	332.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	351.97	66.28	161.18	332.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	252.27	91.89	362.53	1,637.25
Nutrient balance	99.69	-25.61	-201.35	-1,304.36
Applied to removed ratio	1.40	0.72	0.44	0.20

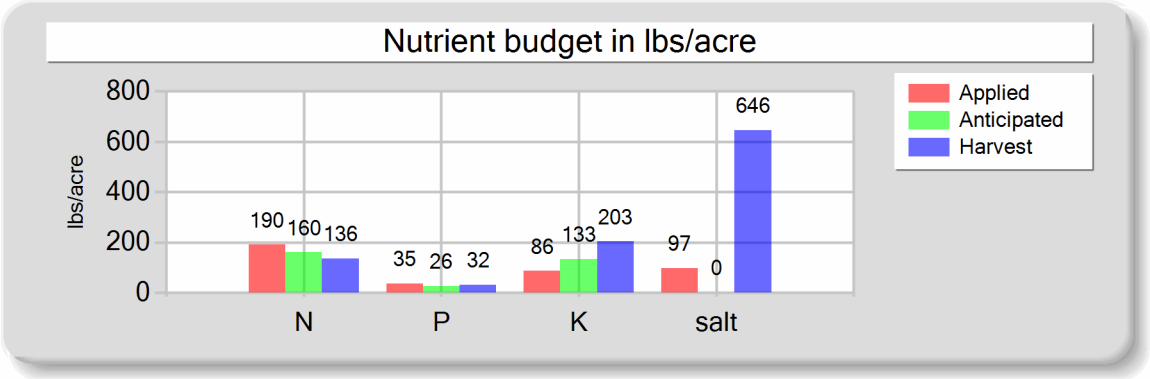
Fresh water applied	
28,493,540.00	*gallons*
1,049.32	*acre-inches*
34.98	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silvera 3 - 11/04/2022: Oats, silage-soft dough

Field name: Silvera 3 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	183.22	35.20	85.61	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	96.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.22	35.20	85.61	96.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	135.51	31.51	203.27	646.05
Nutrient balance	54.71	3.69	-117.66	-549.28
Applied to removed ratio	1.40	1.12	0.42	0.15

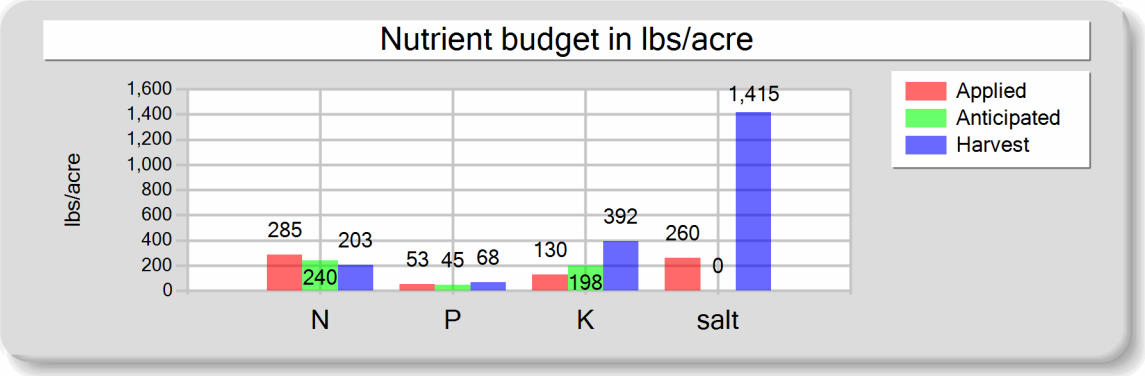
Fresh water applied	
10,767,020.00	*gallons*
396.51	*acre-inches*
10.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silvera 3 - 05/27/2023: Corn, silage

Field name: Silvera 3 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	277.99	53.41	129.89	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	260.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.99	53.41	129.89	260.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	202.93	68.21	392.23	1,415.43
Nutrient balance	82.06	-14.80	-262.33	-1,155.08
Applied to removed ratio	1.40	0.78	0.33	0.18

Fresh water applied	
28,969,480.00	*gallons*
1,066.85	*acre-inches*
27.36	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid manure

Sample and source description: Solid manure

Sample date: 06/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	4,400.00	10,700.00							0.00
DL	0.01	0.01	0.01							0.01

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	800.00	7,000.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 6.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	233.00	84.60	0.00	0.50	39.30	254.00								2,500.00	1,550
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	173.00	134.00	0.00	0.20	35.50	213.00								2,680.00	1,290
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	194.00	182.00	0.00	0.30	38.20	293.00								3,120.00	1,770
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	176.00	153.00	0.00	0.00	35.30	294.00								3,140.00	1,830
DL	1.00	0.50	1.00	1.00	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DW 1

Domestic Well - DW 1

DW1

Sample description: DW1

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.00	0.00	2.00								332.00	
DL	0.10	1.00	0.10								10.00	

Domestic Well - DW 2

DW 2

Sample description: DW 2

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.00	0.00	2.00								328.00	
DL	1.00	1.00	0.10								10.00	

Domestic Well - DW 3

DW 3

Sample description: DW 3

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.00	0.00	2.00								324.00	
DL	0.10	1.00	0.10								10.00	

Irrigation Water - Surface Water

Irrigation Water - Surface Water

Surface Water

Sample description: Surface Water

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								27.10	42
DL	1.00	1.00	1.00								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

26 - 11/20/2020: Clover-grass, hay

26 & 28

Sample and source description: 26 & 28

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,500.00	2,900.00	24,700.00		16.80
DL	100.00	100.00	100.00		0.01

28 - 11/20/2020: Clover-grass, hay

28 - 11/20/2020: Clover-grass, hay

26 & 28

Sample and source description: 26 & 28

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,500.00	2,900.00	24,700.00		16.80
DL	100.00	100.00	100.00		0.01

45 - 12/27/2022: Oats, silage-soft dough

45-A

Sample and source description: 45-A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	3,100.00	20,000.00		5.30
DL	100.00	100.00	100.00		0.01

45-B

Sample and source description: 45-B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	2,900.00	19,700.00		5.70
DL	100.00	100.00	100.00		0.01

45 - 05/11/2023: Corn, silage

45 - 05/11/2023: Corn, silage

45-A

Sample and source description: 45-A

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	2,800.00	19,500.00		7.20
DL	100.00	100.00	100.00		0.01

45-B

Sample and source description: 45-B

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	2,700.00	20,000.00		6.80
DL	100.00	100.00	100.00		0.01

45 - 09/14/2023: Sudangrass, silage

45-A

Sample and source description: 45-A

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,400.00	3,300.00	31,600.00		10.80
DL	100.00	100.00	100.00		0.01

45 - 09/14/2023: Sudangrass, silage

45-B

Sample and source description: 45-B

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 84.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	3,000.00	29,900.00		14.20
DL	100.00	100.00	100.00		0.01

46 - 11/09/2022: Oats, silage-soft dough

46-A

Sample and source description: 46-A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,000.00	17,100.00		8.00
DL	100.00	100.00	100.00		0.01

46-B

Sample and source description: 46-B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	3,000.00	16,900.00		6.30
DL	100.00	100.00	100.00		0.01

46 - 11/09/2022: Oats, silage-soft dough

46-C

Sample and source description: 46-C

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,600.00	18,000.00		9.10
DL	100.00	100.00	100.00		0.01

46 - 05/25/2023: Sudangrass, silage

46-A

Sample and source description: 46-A

Sample date: 07/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	270.00	31,900.00		9.50
DL	100.00	100.00	100.00		0.01

46-B

Sample and source description: 46-B

Sample date: 07/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	330.00	31,900.00		10.01
DL	100.00	100.00	100.00		0.01

46 - 05/25/2023: Sudangrass, silage

46-C

Sample and source description: 46-C

Sample date: 07/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	350.00	27,600.00		9.40
DL	100.00	100.00	100.00		0.01

47 - 11/09/2022: Oats, silage-soft dough

47-A

Sample and source description: 47-A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	4,000.00	15,200.00		7.20
DL	100.00	100.00	100.00		0.01

47-B

Sample and source description: 47-B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	4,000.00	15,300.00		7.10
DL	100.00	100.00	100.00		0.01

47 - 05/25/2023: Sudangrass, silage

47 - 05/25/2023: Sudangrass, silage

47-A

Sample and source description: 47-A

Sample date: 07/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	360.00	40,900.00		11.80
DL	100.00	100.00	100.00		0.01

47-B

Sample and source description: 47-B

Sample date: 07/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	350.00	17,000.00		9.00
DL	100.00	100.00	100.00		0.01

Carson - 11/09/2022: Oats, silage-soft dough

Carson

Sample and source description: Carson

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,300.00	15,600.00		5.60
DL	100.00	100.00	100.00		0.01

Carson - 05/27/2023: Corn, silage

Carson - 05/27/2023: Corn, silage

Carson

Sample and source description: Carson

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,900.00	2,600.00	14,400.00		6.60
DL	100.00	100.00	100.00		0.01

CMI - 11/04/2022: Oats, silage-soft dough

CMI

Sample and source description: CMI

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	4,500.00	23,900.00		8.60
DL	100.00	100.00	100.00		0.01

Collins - 10/24/2022: Oats, silage-soft dough

Collins A

Sample and source description: Collins A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	3,400.00	16,300.00		5.50
DL	100.00	100.00	100.00		0.01

Collins - 10/24/2022: Oats, silage-soft dough

Collins B

Sample and source description: Collins B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,700.00	3,700.00	18,100.00		5.70
DL	100.00	100.00	100.00		0.01

Collins - 05/10/2023: Corn, silage

Collins A

Sample and source description: Collins A

Sample date: 08/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	3,400.00	17,300.00		7.70
DL	100.00	100.00	100.00		0.01

Collins B

Sample and source description: Collins B

Sample date: 08/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	4,100.00	21,400.00		8.20
DL	100.00	100.00	100.00		0.01

Field 44 - 11/20/2020: Clover-grass, hay

Field 44 - 11/20/2020: Clover-grass, hay

44

Sample and source description: 44

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,500.00	3,200.00	26,100.00		25.10
DL	100.00	100.00	100.00		0.01

Silvera 1 - 11/04/2022: Oats, silage-soft dough

Silvera 1 - A

Sample and source description: Silvera 1 - A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	5,300.00	29,300.00		9.00
DL	100.00	100.00	100.00		0.01

Silvera 1 - B

Sample and source description: Silvera 1 - B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	4,400.00	24,600.00		8.10
DL	100.00	100.00	100.00		0.01

Silvera 1 - 05/10/2023: Corn, silage

Silvera 1 - 05/10/2023: Corn, silage

Silveira 1 - A

Sample and source description: Silveira 1 - A

Sample date: 08/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	4,300.00	22,800.00		10.30
DL	100.00	100.00	100.00		0.01

Silveira 1 - B

Sample and source description: Silveira 1 - B

Sample date: 08/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	4,400.00	20,000.00		9.30
DL	100.00	100.00	100.00		0.01

Silvera 2 - 11/04/2022: Oats, silage-soft dough

Silvera 2

Sample and source description: Silvera 2

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	3,600.00	23,400.00		7.60
DL	100.00	100.00	100.00		0.01

Silvera 2 - 05/10/2023: Corn, silage

Silvera 2 - 05/10/2023: Corn, silage

Silveira 2

Sample and source description: Silveira 2

Sample date: 08/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	5,500.00	21,700.00		9.80
DL	100.00	100.00	100.00		0.01

Silvera 3 - 11/04/2022: Oats, silage-soft dough

Silvera 3

Sample and source description: Silvera 3

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	4,000.00	25,800.00		8.20
DL	100.00	100.00	100.00		0.01

Silvera 3 - 05/27/2023: Corn, silage

Silvera 3

Sample and source description: Silvera 3

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	4,000.00	23,000.00		8.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

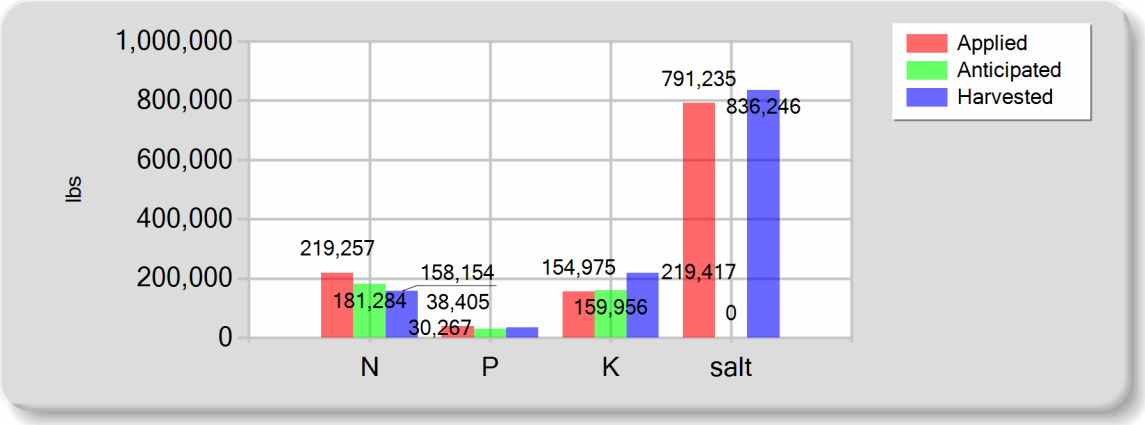
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

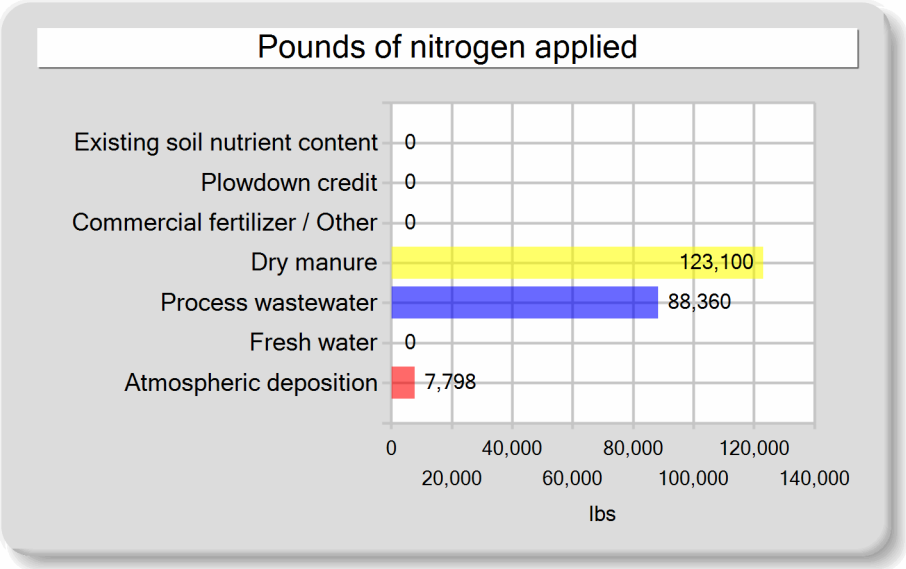
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

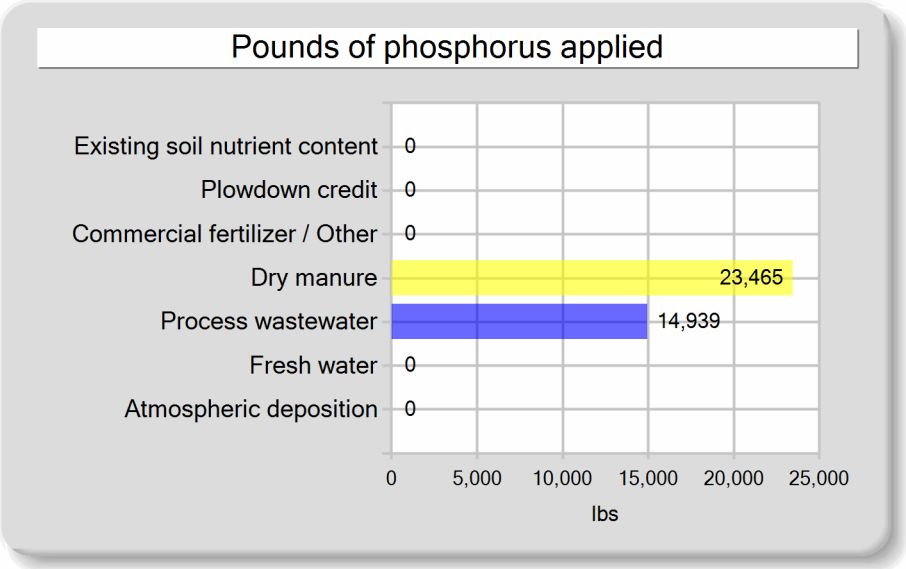
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	123,099.57	23,465.39	57,852.08	0.00
Process wastewater	88,359.70	14,939.39	97,122.78	592,456.78
Fresh water	0.00	0.00	0.00	198,778.47
Atmospheric deposition	7,798.00	0.00	0.00	0.00
Total nutrients applied	219,257.27	38,404.78	154,974.86	791,235.25
Anticipated crop nutrient removal	181,284.00	30,267.00	159,955.60	0.00
Actual crop nutrient removal	158,154.16	34,079.14	219,416.74	836,245.90
Nutrient balance	61,103.10	4,325.64	-64,441.88	-45,010.65
Applied to removed ratio	1.39	1.13	0.71	0.95

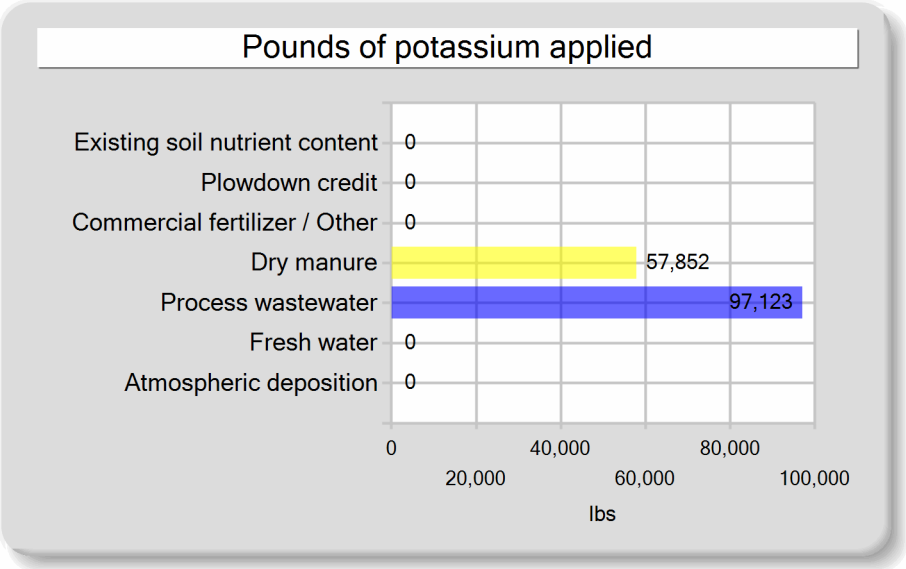
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

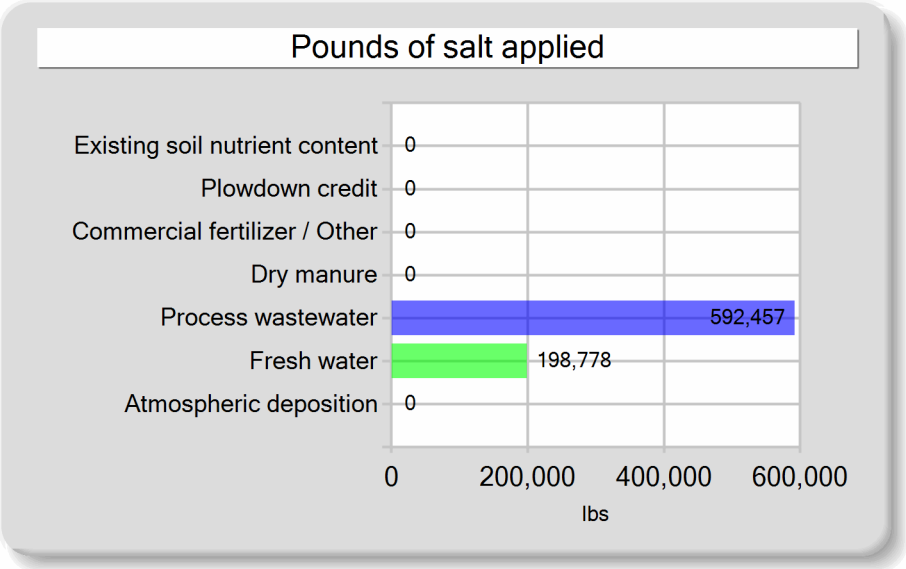


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes
Lloyd Fagundes (Jun 25, 2024 17:06 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Fagundes Bros. Dairy C/O Ralph, Lloyd, &/or Fred Fagundes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/25/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #3

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:32

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0561-01	DW 1	Ag Water	T. Madruga		08/02/2023 16:00
23H0561-02	DW 2	Ag Water	T. Madruga		08/02/2023 16:00
23H0561-03	DW 3	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -1.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #3

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:32

Sample Results

Sample: DW 1
23H0561-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.33**	mmhos/cm	0.01	1		08/04/23 14:41	SM 2510 B		BEH0246
Electrical Conductivity umhos	**332**	umhos/cm	10.0	1		08/04/23 14:41	SM 2510 B		BEH0246
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0304
Nitrate Nitrogen as NO3N	**2.0**	mg/L	0.1	1	10	08/03/23 19:51	EPA 300.0		BEH0186
pH	**7.9**	units	1.0	1		08/04/23 14:41	SM 4500-H+	H	BEH0246
Temperature	**25.0**	°C	0.0	1		08/04/23 14:41	SM 2510 B		BEH0246

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R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #3

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:32

Sample Results
(Continued)

Sample: DW 2
23H0561-02 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.33**	mmhos/cm	0.01	1		08/04/23 14:43	SM 2510 B		BEH0246
Electrical Conductivity umhos	**328**	umhos/cm	10.0	1		08/04/23 14:43	SM 2510 B		BEH0246
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0304
Nitrate Nitrogen as NO3N	**2.0**	mg/L	0.1	1	10	08/03/23 20:12	EPA 300.0		BEH0186
pH	**8.0**	units	1.0	1		08/04/23 14:43	SM 4500-H+	H	BEH0246
Temperature	**25.0**	°C	0.0	1		08/04/23 14:43	SM 2510 B		BEH0246

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #3

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:32

Sample Results
(Continued)

Sample: DW 3
23H0561-03 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.32**	mmhos/cm	0.01	1		08/04/23 14:44	SM 2510 B		BEH0246
Electrical Conductivity umhos	**324**	umhos/cm	10.0	1		08/04/23 14:44	SM 2510 B		BEH0246
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0304
Nitrate Nitrogen as NO3N	**2.0**	mg/L	0.1	1	10	08/03/23 20:32	EPA 300.0		BEH0186
pH	**8.0**	units	1.0	1		08/04/23 14:44	SM 4500-H+	H	BEH0246
Temperature	**25.0**	°C	0.0	1		08/04/23 14:44	SM 2510 B		BEH0246

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #3

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:32

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0186										
Blank (BEH0186-BLK1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0186-BLK2)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0186-BS1)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		93.5	90-110		
Duplicate (BEH0186-DUP1)		**Source: 23H0558-01**			Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	0.07		0.1	mg/L		0.07			1.48	10
Matrix Spike (BEH0186-MS1)		**Source: 23H0558-01**			Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.07	92.2	90-110		
Reference (BEH0186-SRM1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.7	90-110		

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #3

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0246										
Blank (BEH0246-BLK1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0246-BLK2)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0246-BLK3)					Prepared & Analyzed: 8/4/2023					
pH	7.7		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEH0246-DUP1)		**Source: 23H0547-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.03		0.01	mmhos/cm		0.03			8.26	10
pH	7.8		1.0	units		7.8			0.383	10
Electrical Conductivity umhos	26.7		10.0	umhos/cm		29.0			8.26	10
Duplicate (BEH0246-DUP2)		**Source: 23H0561-03**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.33		0.01	mmhos/cm		0.32			0.277	10
pH	8.0		1.0	units		8.0			0.251	10
Electrical Conductivity umhos	325		10.0	umhos/cm		324			0.277	10
Reference (BEH0246-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	547			umhos/cm	538.0		102	90-110		
Reference (BEH0246-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.8			units	7.790		101	.7163-101.28		
Reference (BEH0246-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0246-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0246-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #3

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:32

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0246 (Continued)										
Reference (BEH0246-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0246-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0246-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0246-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 3 No. of Bottles: 3

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Fagundes Brothers Dairy #3
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) **DWW1: EC, pH, NO_3-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW 1	8/2/2023	1600	-1.6	
2	DW 2	8/2/2023	1600	-1.4	
3	DW 3	8/2/2023	1600	0.0	
4					
5					
6	IR Thermometer SN: 221314357				
7	Correction Factor: 0°C				
8	Calibration Due: 9/26/2023 Location: Laboratory				
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JLF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23 15:10	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:10 23H0561

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										